

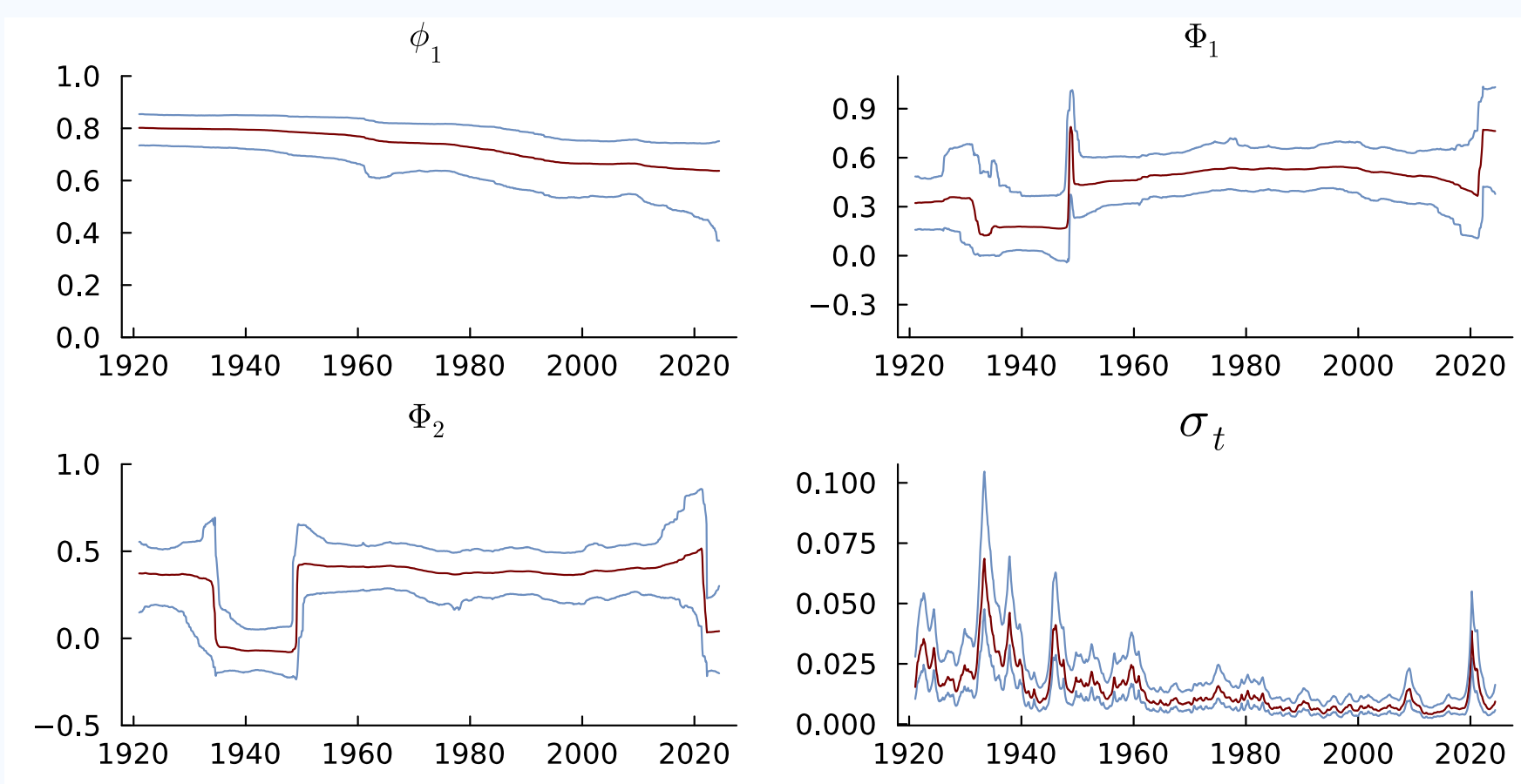
Bayesian Time-Varying Multi-Seasonal ARMA Models

Stable and Scalable Inference for Nonstationary Time Series

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1. Why TV-SARMA Models?

- Smooth evolution and abrupt shifts
- Multiple seasonalities (daily, weekly, annual)
- Adaptive models



TV-SAR(1,2)₁₂: posterior median and 95% credible bands.

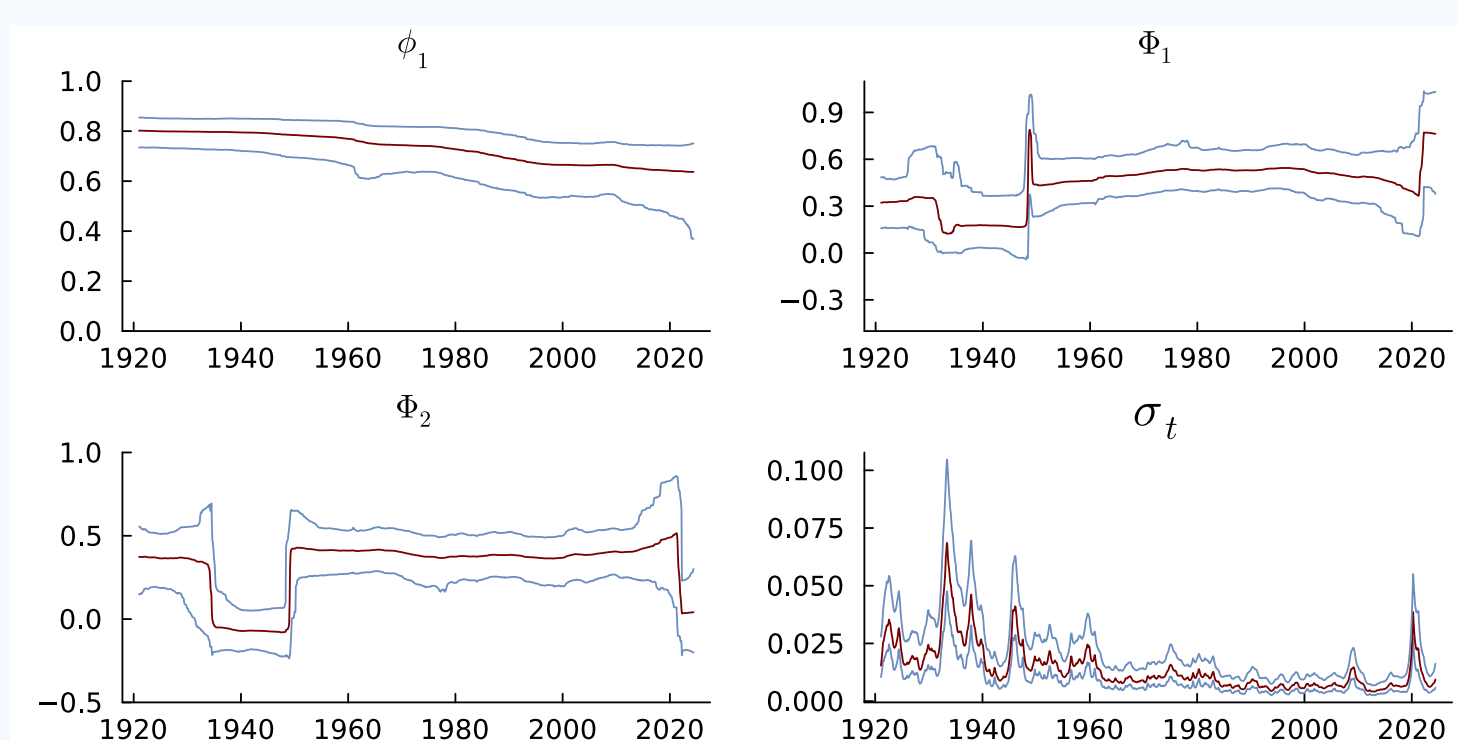
2. Modeling framework/contributions

- Bayesian TV-SARMA
- Multiple seasonalities
- Nonlinear state-space inference
- Regularization with DSP priors
- Stable and scalable inference

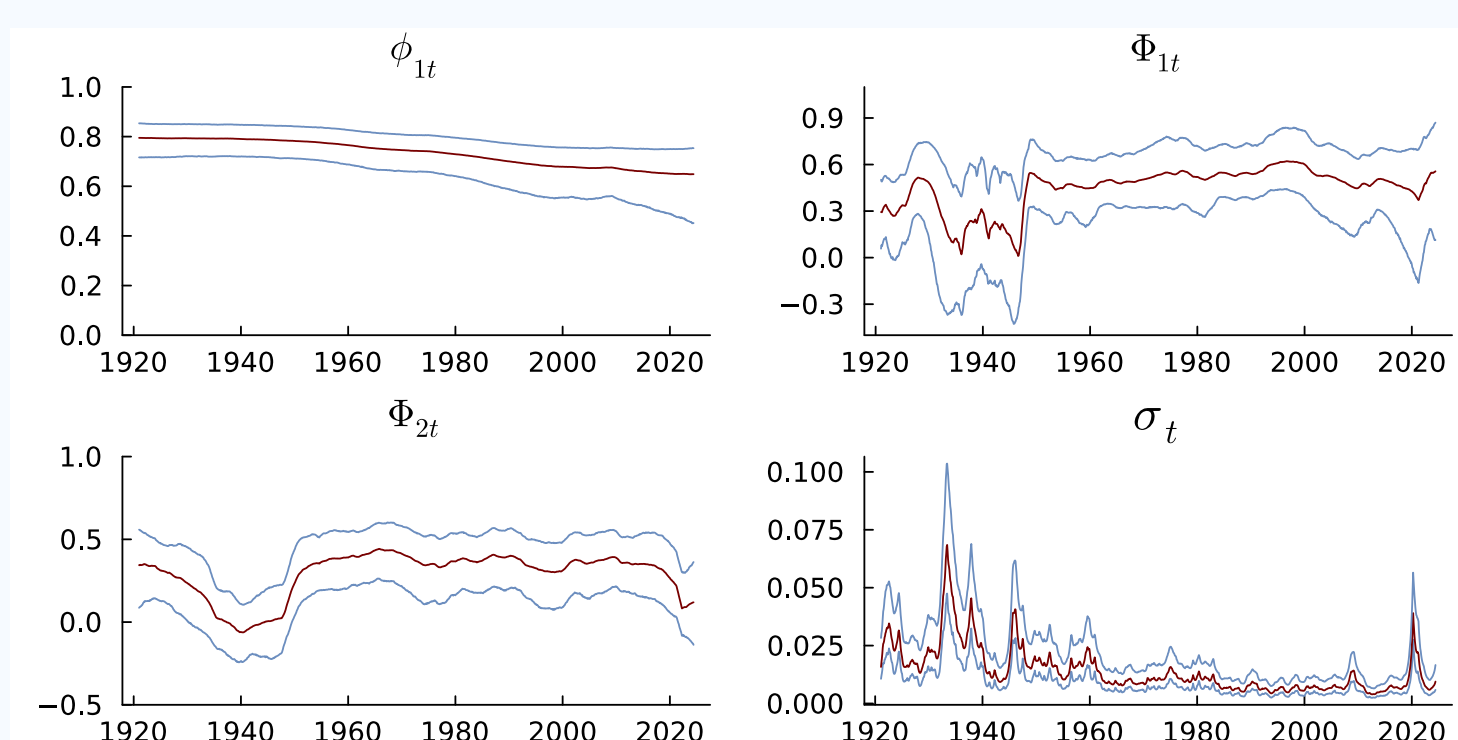
3. State-space representation

$$y_t = \mathbf{x}_{0t}^\top \tilde{\phi}_t^{(AR)} + \mathbf{z}_{0t}^\top \tilde{\psi}_t^{(MA)} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma^2)$$
$$\tilde{\phi}_t, \tilde{\psi}_t = \tilde{\mathbf{g}}(\theta_t) \quad \leftarrow \text{stability/invertibility constraint}$$
$$\theta_t = \theta_{t-1} + \nu_t, \quad \nu_t \sim N(0, \text{Diag}(\exp(\mathbf{h}_t))) \quad \leftarrow \text{DSP prior}$$
$$\mathbf{h}_t = \mu + \kappa(\mathbf{h}_{t-1} - \mu) + \eta_t, \quad \eta_t \sim Z(1/2, 1/2, 0, 1).$$

4. DSP vs. Gaussian prior

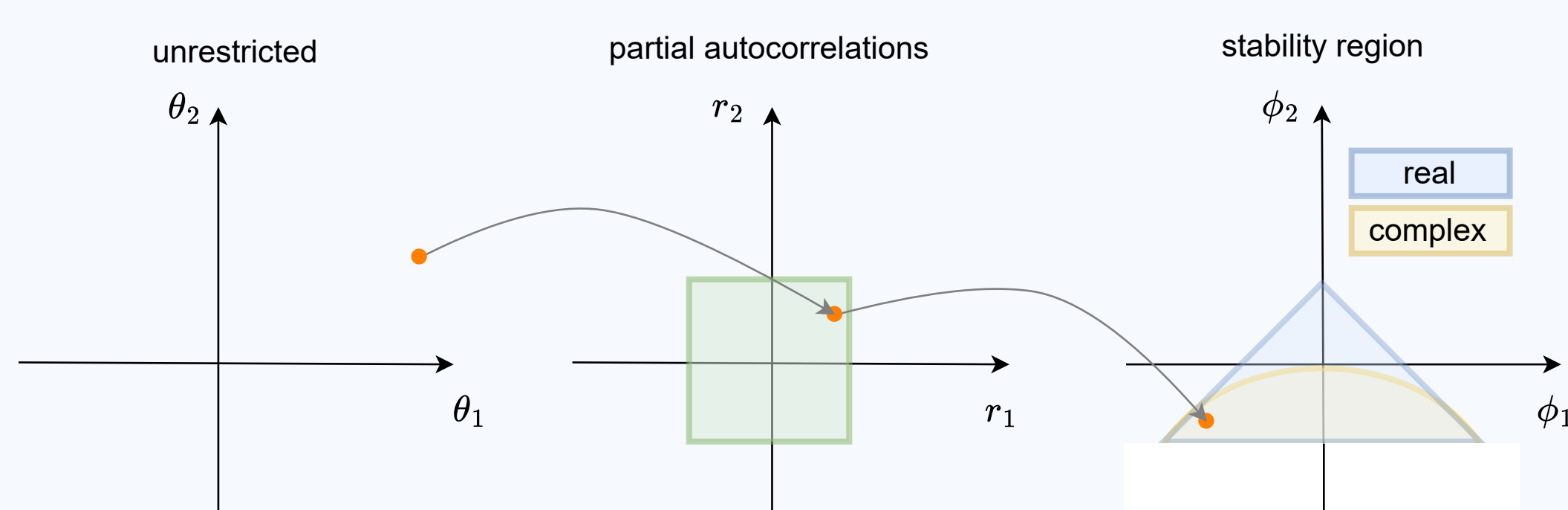


DSP prior with SV



Gaussian prior with SV

5. Stability parametrization and uniform prior

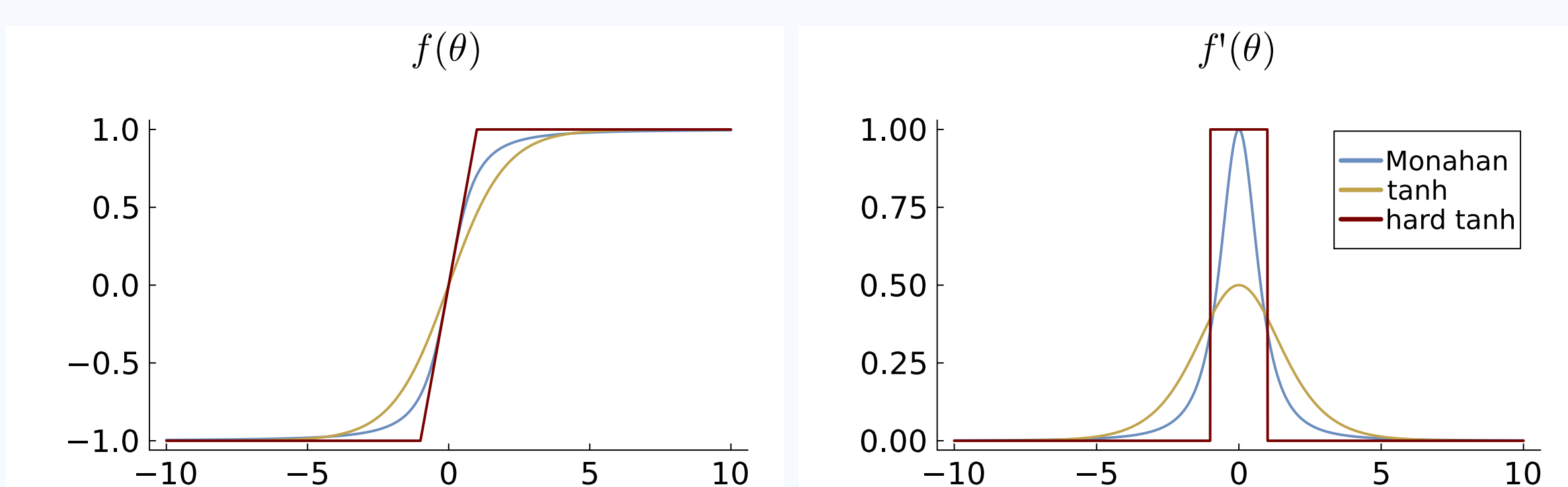


Monahan transformation + uniform prior over AR parameter space

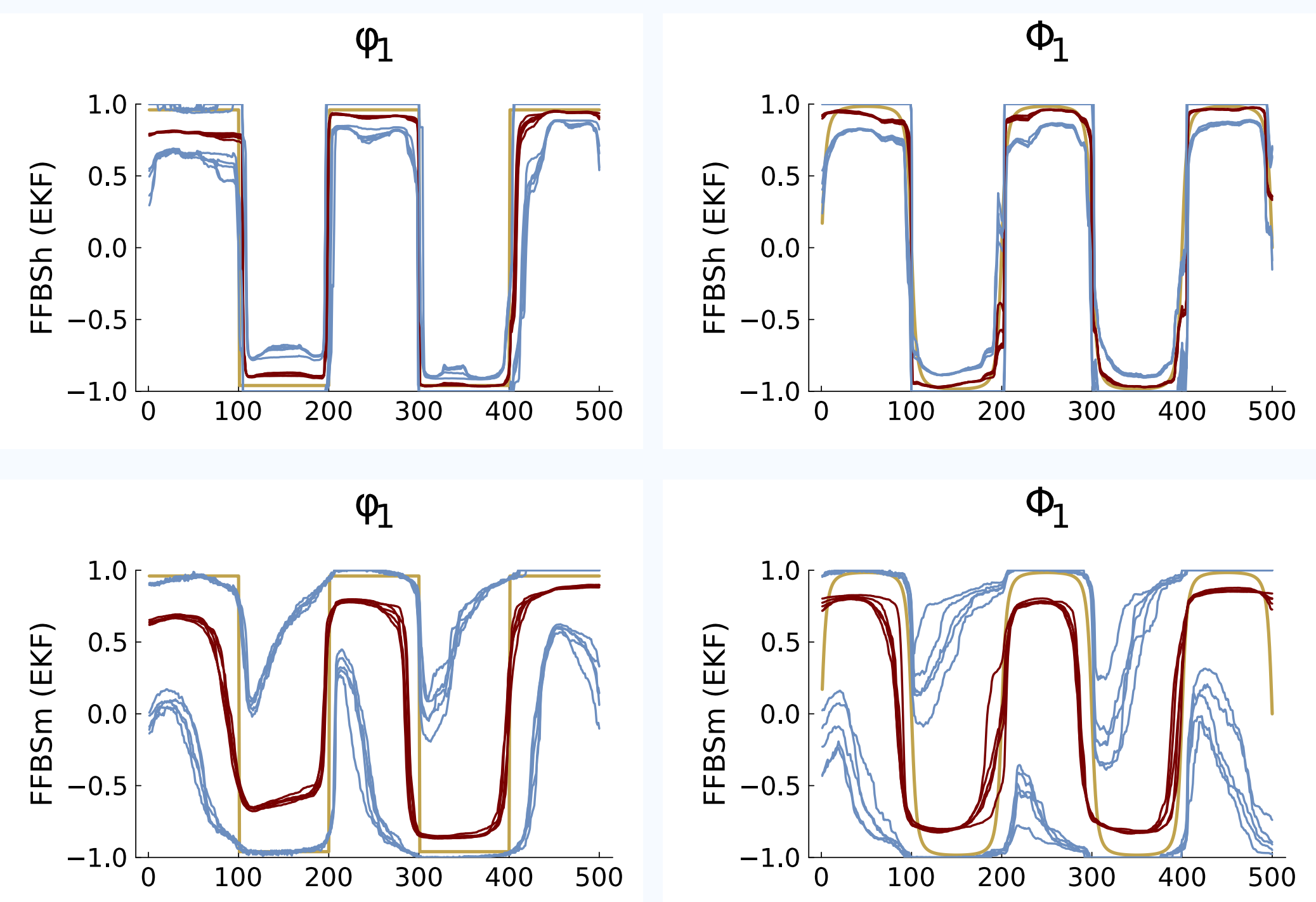
(Same idea applies to ensure invertibility!)

6. Stability parametrization at the boundary

Monahan (1984) $\Rightarrow$ $\tanh \Rightarrow$ **hardtanh**



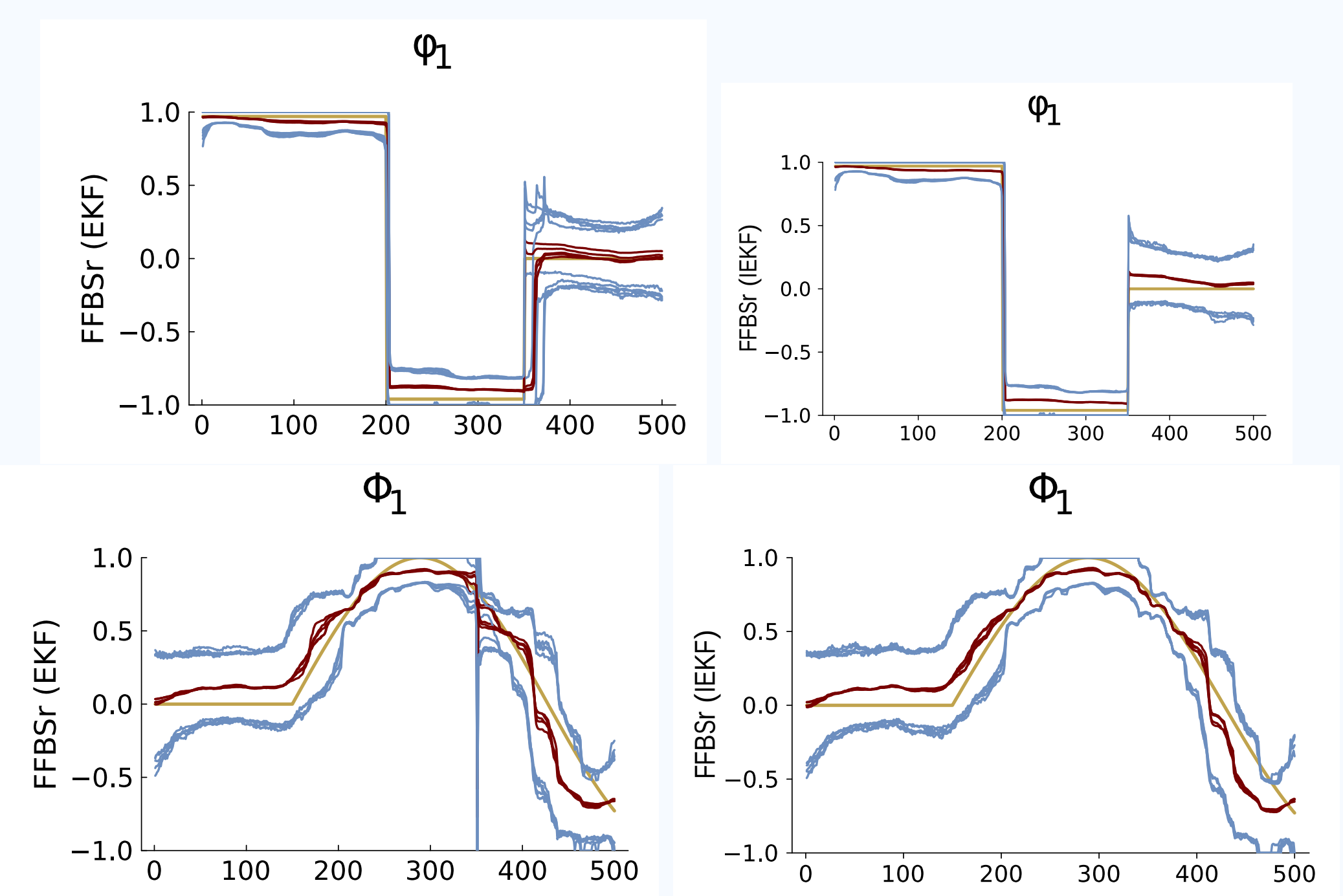
7. Parametrization effect on inference



TV-SAR(1,1)₁₂: **Hardtanh** (first row) and **Monahan** (second row)

8. Adaptive inference

FFBS+EKF $\xrightarrow{\text{NIS trigger}}$ **FFBS+IEKF**



TV-SAR(1,1)₁₂: Baseline (first row) and Adaptive (second row)

Fast when possible, accurate when necessary!

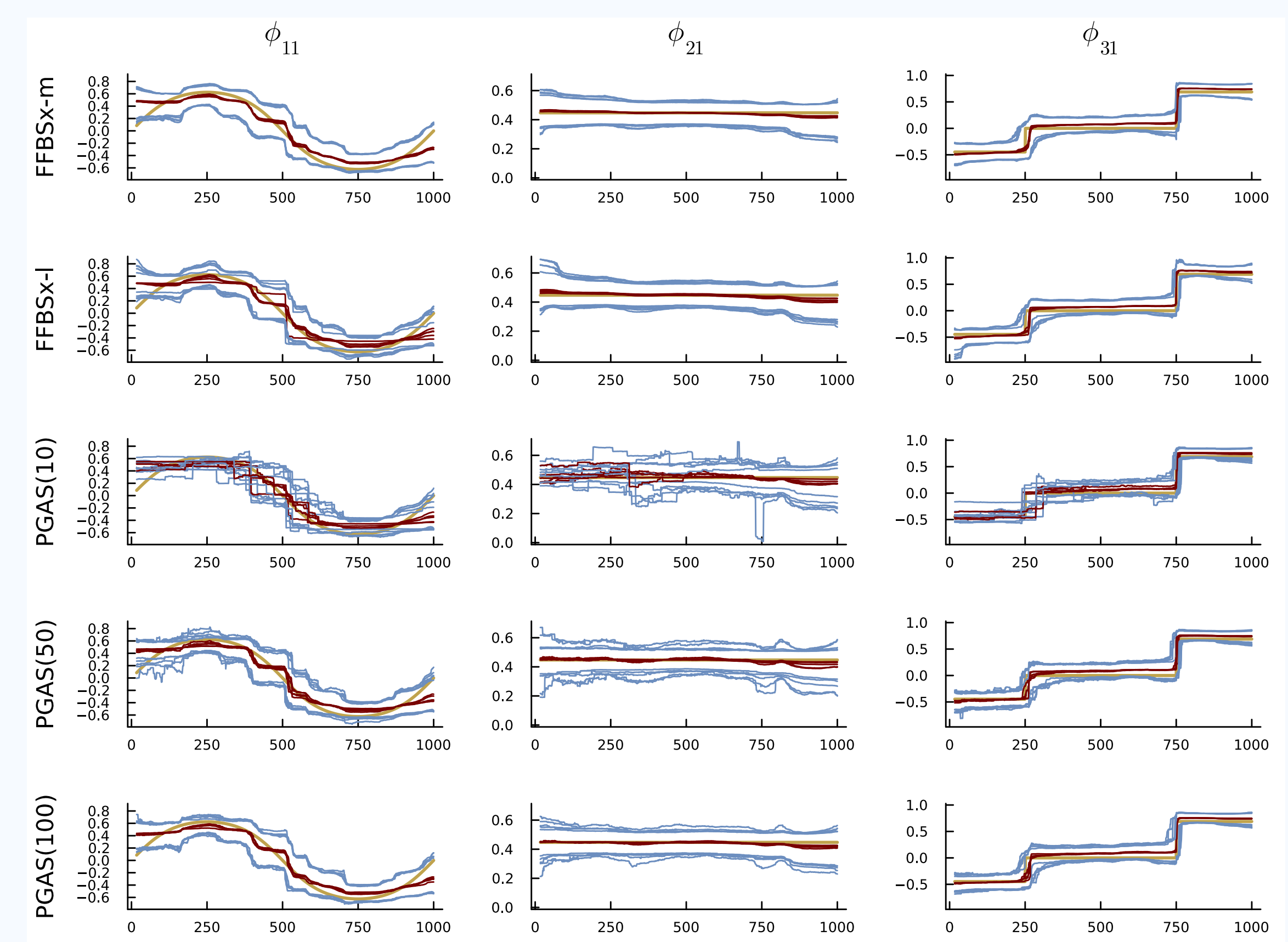
9. Gibbs sampling scheme

States $\rightarrow$ DSP $\rightarrow$ ARMA $\rightarrow$ Presample $\rightarrow$ Errors $\rightarrow$ Volatility

10. Scaling to multiple seasonalities

Grouping $\Rightarrow$ Sequential EKF $\Rightarrow$ **Scalable Inference**
(Fewer state updates) (Lower filtering cost) (Long high-frequency series)

11. Simulations



TV-SAR(1,1,1)_{4,12}: Posterior median (red) and 95% credible bands (blue)



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